

Enterprise Digital Infrastructure

Monitoring



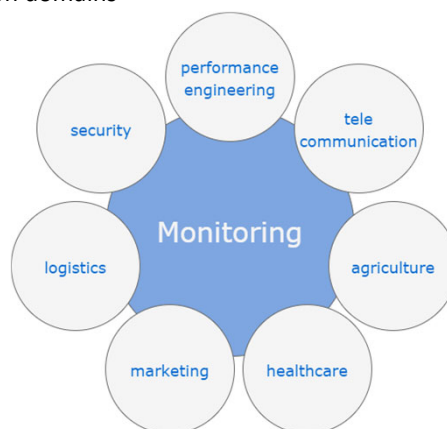
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Monitoring

Monitoring, i.e., the process of collecting measurements, is a *key activity* in many application domains



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Monitoring

Monitoring is fundamental to properly manage infrastructures and services

Main purposes of monitoring activities:

- Understand the behavior and performance of services and infrastructures
- Assess their health
- Anticipate potential outages and SLA violations
- Diagnose performance issues when something goes wrong
- Proactively address and mitigate performance issues
- Investigate user behaviors

Data being monitored is of interest of different stakeholders, e.g., end users, software developers, webmasters, service providers, network operators, cloud operators

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Basic concepts

Monitoring denotes the *process* of measuring “something” by means of a *device* or a measurement instrument/tool in a standard *measurement unit*

The result of the process is a number or a set of numbers, i.e., *measurements*, that provide a *quantitative* assessment of the measured object

Measurements must be specific of the components under investigation

The *accuracy* of the process and the *precision* of the measurement tools are *necessary* ingredients for the validity of the measurements, although not *sufficient*

A correct analysis and interpretation of the measurements are required before drawing any meaningful conclusion (e.g., *domain knowledge*, data science competences)

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Monitoring activities

Prerequisites of any monitoring activity:

- Definition of *clear objectives*
- *Solid knowledge* of the domain being monitored

The objective is to learn how PLAN, DESIGN & SETUP a monitoring project

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Monitoring project: methodological approach

The plan, design and setup of a monitoring project are formulated as a workflow consisting of multiple *interdependent* steps

Each step refers to a specific aspect of the project:

- WHY: definition of the *objectives* of the project
- WHICH/WHO: definition of the *target(s)* of the project
- WHAT: definition of the *attributes* to be measured
- WHERE: definition of the *vantage point(s)*
- How: definition of the *measurement process*
- WHEN: definition of the *timing*

Once measurements have been collected, they need to be *summarized* to build *models* representative of the properties/behavior of the target

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Recap

Monitoring systems and services provides quantitative assessments useful for understanding their behavior

The setup of a monitoring project is based on a methodology driven by various questions

Monitoring requires multidisciplinary competences

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WHY

This step refers to the definition of the objective(s) of the monitoring project

Possible objectives:

- to understand how a technological infrastructure/service actually work
- to discover bugs (*debugging*)
- to detect/diagnose inefficiencies (*performance debugging*)
- to check status (*predict failures*)
- to measure the usage of infrastructure/services (*accounting*)
- to detect anomalies (*security*)
- to understand the behavior/activities of the users (*profiling*)

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WHICH/WHO

This step refers to the definition of the *target* of the monitoring project

The choice depends on the objectives of the project (i.e., why)

Possible targets:

- *Technological infrastructure* at different levels (not to be considered in isolation)
 - computing infrastructure
 - storage infrastructure
 - communication infrastructure
 - software components
- *Workload*
- *Network traffic*
- *Users*

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WHAT

This step refers to the definition of the *attributes*, i.e., properties of the target

These choices also depend on the usage of the measurements

- *Offline*
- *Online/on the fly*

Possible attributes:

- number of processes, RAM usage, CPU usage, number/types of instructions, number of cache misses, number of I/O operations
- application processing time, number of DNS queries, number/type of DB transactions, number/types of HTTP transactions, user navigation profile
- throughput, latency, jitter, error rate, number of dropped packets, protocol types
- GPS coordinates, energy, battery usage, temperature

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WHERE

This step refers to the definition of the *vantage point(s)*, i.e., location(s), where measurements about the target are collected

Any component of an infrastructure can be chosen as a vantage point

The choice has to take into account two important aspects: *feasibility* AND *impact*

Feasibility refers to the *permissions/authorizations* (e.g., GDPR) required to monitor technological infrastructures and services

Impact refers to *side effects* such as *perturbations*, *intrusiveness* and *overhead* introduced by monitoring activities (e.g., monitoring frequency, length of monitoring interval, amount of collected data)

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HOW

This step refers to the definition of the *measurement process* that includes the choice of the *techniques* and *tools* to be used to collect measurements

The choice is between two categories:

- *Active* techniques aimed at collecting measurements on the target component under a *controlled* workload/traffic

artificial workload/traffic & real infrastructure

- *Passive* techniques aimed at collecting measurements on the target component while operating under its *real* workload/traffic

real workload/traffic & real infrastructure

The measurement process usually leverages (properly customized) open source or proprietary *tools*

These tools can be used in isolation or in combination

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WHEN

This step refers to the definition of the timing of the monitoring project, i.e., when measurements are to be collected

Possible choices:

- Continuously
- Periodically
- Occasionally

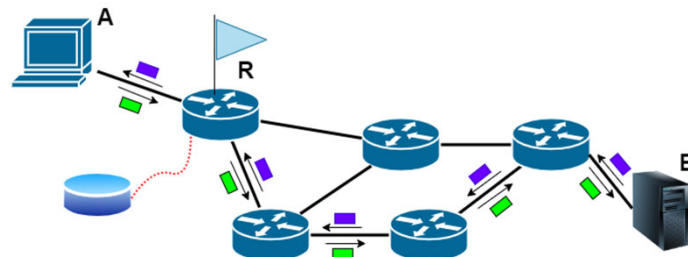
In general, it is recommended to consider monitoring as a *systematic* activity rather than a sporadic activity

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Example of monitoring scenarios



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Active monitoring

Active monitoring belongs to the broad category of *load* and *traffic generators*

It is aimed at analyzing the properties and assessing the efficiency of a real infrastructure under a “controlled” (artificial) workload/traffic

Main phases:

- definition of the characteristics of the artificial load/traffic (i.e., *benchmark*)
- generation of artificial load/traffic on the real infrastructure chosen as target
- monitoring the performance/behavior of the target

The complexity of the generators varies from simple commands/utilities that are part of the operating systems, to a large variety of speed testing tools and complex benchmarking frameworks

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Active monitoring (cont'd)

The target (i.e., System Under Test SUT) can be *shared* or *dedicated*

The artificial load is generated according to some predefined characteristics/behaviors

To increase accuracy of the experiments and ensure measurement quality, it is necessary to

1. Generate a *realistic/representative load* (consisting of multiple requests/packets)
2. Repeat the experiments *multiple times* and discard inaccurate samples

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Monitoring project

Setup a monitoring project aimed at diagnosing the health of a communication infrastructure

Simple commands for active monitoring:

- **ping**
- **tracert**
- **mtr**
- **hping3**
- **nmap**
- ...

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Active monitoring: **ping**

ping is a simple active monitoring tool aimed at managing, measuring, testing and debugging network configurations

The tool relies on the Internet Control Message Protocol (ICMP), a network layer protocol aimed at error reporting and diagnosis [RFC 792 released in 1981]

ICMP supports 11 types of messages sent as IP datagrams

Ping generates packets from a sender (i.e., *vantage point*) towards a *target*

Each packet consists of ICMP and IP headers (8+20 bytes) and variable payloads (pad bytes to fill the packet)

Ping is based on two ICMP messages: ECHO msg (type 8) that elicits an ECHO REPLY message (type 0)

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Active monitoring: **ping** scenario

Why:

- Debugging network configuration

Which:

- Server B

What:

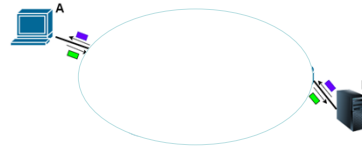
- Reachability of the target
- Instantaneous RTT between vantage point and target
- Packet loss rate

Where:

- Computer A

How:

- **ping**



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Active monitoring: **ping** limitations

There are several *limitations* affecting the accuracy of the measurements:

- Impossible to detect whether/where there is a network congestion
- Impossible to understand why the *target* is *unreachable* (e.g., network “broken”, target *down*, intermediate routers *discarding* packets)

To ensure measurement quality, it is important to *repeat* the tests multiple times especially in the case of *shared* media

These tests should **not** be performed during normal operations, e.g., working hours, or using automated scripts because of the traffic being generated (e.g., flooding)

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Active monitoring: **traceroute**

traceroute is a simple active monitoring tool aimed at discovering the network topology between a vantage point and a target

The tool utilizes the Time To Live (TTL) header field of IP packets

The traffic generated from the vantage point consists of (multiple) *independent* sequences of *independent* packets (by default three) towards the target (implementations based on ICMP, TCP or UDP protocols)

Each sequence is characterized by a different value of TTL (i.e., 1, 2, ..., N)

A TTL equal to zero elicits an ICMP `TIME_EXCEEDED` response from the router along the path to the target back to the vantage point

The target generates an ICMP *"port unreachable"* response

Target and intermediate routers are passive entities

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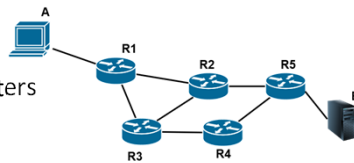
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Active monitoring: **traceroute** scenario

Why:

- Discover network topology, loops, misconfigurations, malfunctioning routers



Which:

- Server B

What:

- Path between vantage point and target
- Instantaneous RTTs between vantage point and all intermediate routers
- Instantaneous RTTs between vantage point and target

Where:

- Computer A

How:

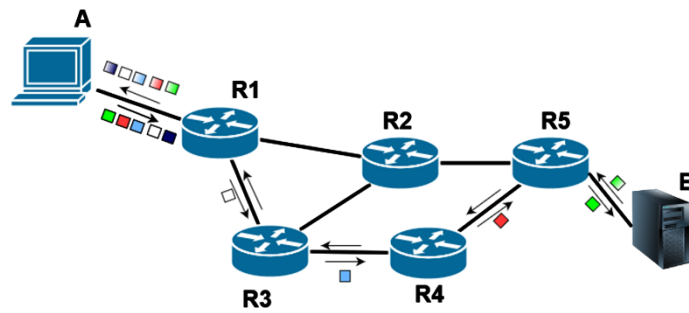
- **traceroute**

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Implementation



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Active monitoring: **traceroute** limitations

There are several *limitations* affecting the accuracy of the measurements

- Asymmetric paths due to network congestion
- Unstable paths due to temporary failures
- Blocked paths due to disconnected networks or routers discarding packets

Packets are independent entities that might follow different paths

It is important to *repeat* the tests multiple times and discard inaccurate measurements to avoid drawing wrong conclusions

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Stress test tools

Stress test tools are load/traffic generators aimed at measuring the (maximum) performance of services and infrastructure under a given load

Tools can be used for malicious purposes (e.g., stressers/booters used by DDoS-for-hire services)

Speed testing tools are a specific type of stress test tools

Speed testing tools aim at measuring the network speed by generating *artificial* traffic towards a server chosen as a target

Typical “black-box” approach

The traffic consists of downloads, uploads, pings and ...

In general, measurements being collected refer to downstream and upstream throughput, latency, jitter, at a given instant of time

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Speed testing tool limitations

Speed testing tools are affected by several limitations:

- Measurements are influenced by many factors, e.g., testing infrastructure, user hardware/software, “distance” between target and vantage point, network conditions
- Artificial traffic being generated is not always representative of user’s Internet experience

To improve the accuracy of the measurements, it is necessary to repeat the tests multiple times at different times of the day and in different days of the week

It is also important to vary the target

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Ookla Speedtest®



11+ million
tests daily



Billions
of daily samples



15,000+
testing servers in 190+
countries



100%
of the world's
countries represented
in Ookla data



55+ billion
tests to date

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Ookla collects certain data through Speedtest that may be considered personally identifiable, such as your IP address, unique device identifiers or location. Ookla believes it has a legitimate interest to share this data with internet providers, hardware manufacturers and industry regulators to help them understand and create a better and faster internet. For further information including how the data may be shared, where the data may be transferred and Ookla's contact details, please see our [Privacy Policy](#).

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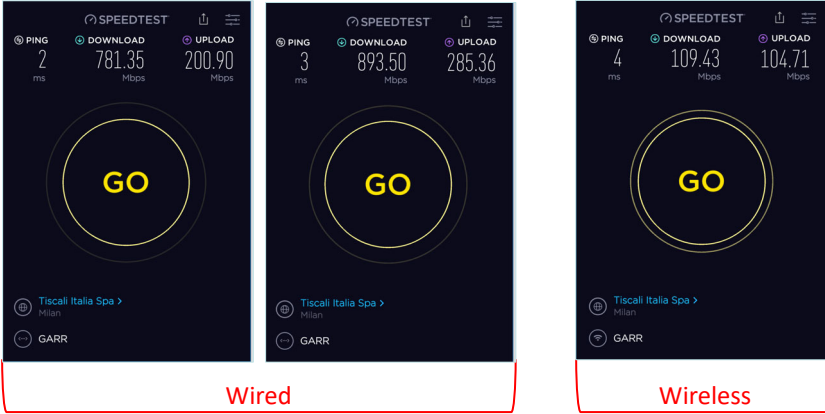
Information collected through Speedtest and Sensorly [may not be anonymous](#) as follows: for some ISPs and mobile carriers, Ookla will provide full IP addresses and/or network, hardware or device identifiers for such ISP or mobile carrier's customers upon request. Ookla may also provide full IP addresses and/or network, hardware and device identifiers, to equipment manufacturers and core network, or wholesale providers on a case-by-case basis for network research purposes. Ookla may also collect [GeoIP Data and/or Precise Location Data](#), and such data may also be licensed and/or transferred to third parties.

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Ookla Speedtest® : some results

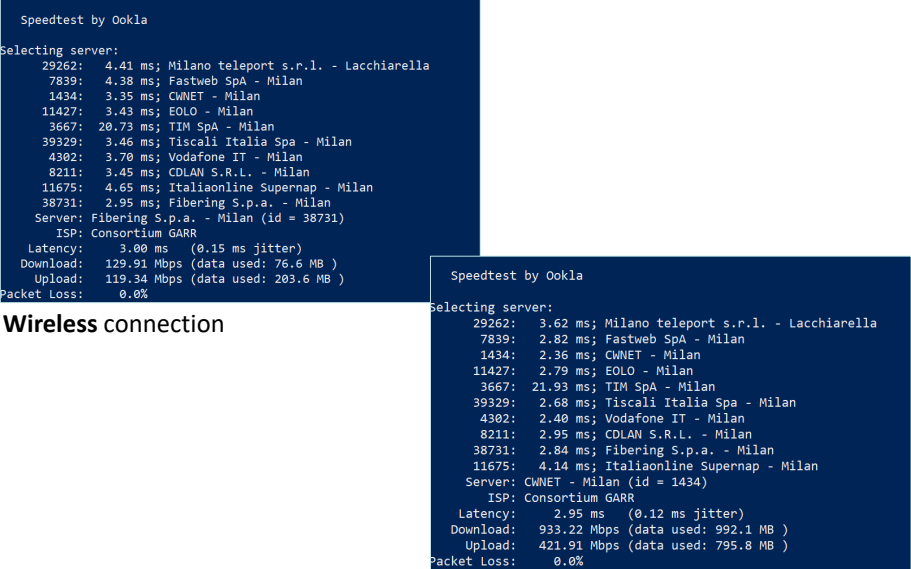


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Other Speedtest® results on CLI



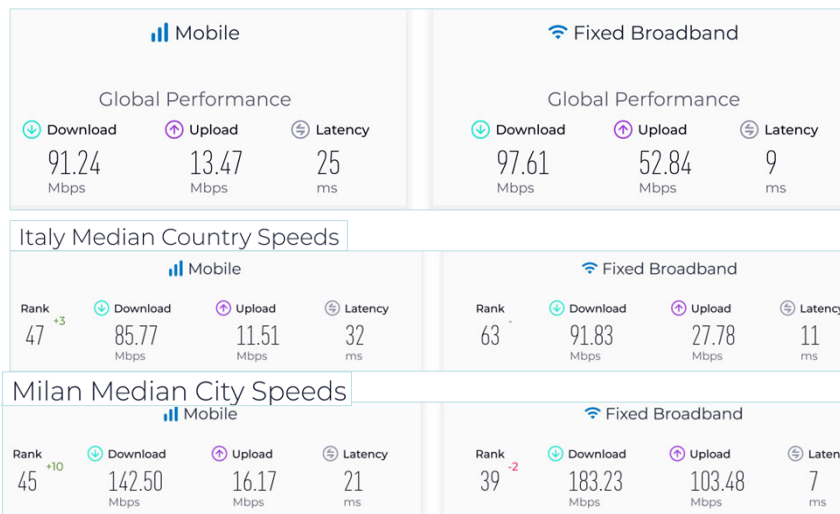
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Wired connection

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Speedtest Global Index [January 2025]



[Speedtest Global Index](#)

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Speedtest Global Index: Countries [Jan 2025]

#	Country	Mbps	#	Country	Mbps
1	United Arab Emirates	545.94	1	Singapore	336.45
2	Qatar	521.51	2	United Arab Emirates	310.05
3	Kuwait	314.39	3	Hong Kong (SAR)	305.71
4	Bulgaria	252.05	4	France	287.44
5	Bahrain	213.09	5	Iceland	281.95
6	Brazil	210.38	6	Chile	276.77
7	Saudi Arabia	201.08	7	United States	274.16
8	Denmark	198.33	8	Denmark	247.62
9	South Korea	195.27	9	Spain	245.58
10	Georgia	184.56	10	Switzerland	242.32
11	Malaysia	171.61	11	Romania	238.22
12	China	171.01	12	China	238.04

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Speedtest Global Index: Cities [Jan 2025]

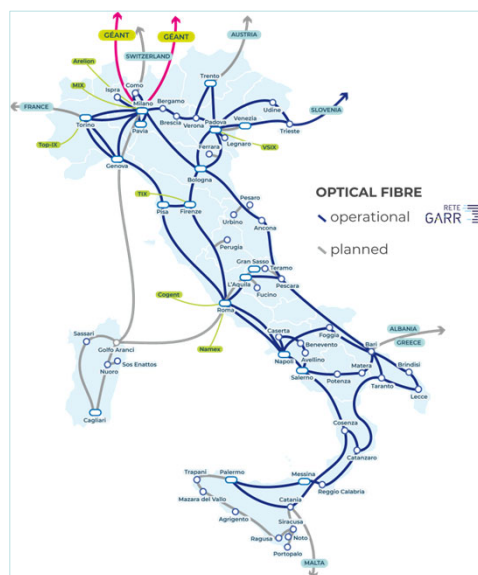
#	City	Mbps	#	City	Mbps
1	- Abu Dhabi, United Arab Emirates	597.74	1	+2 Abu Dhabi, United Arab Emirates	331.46
2	+1 Ar-Rayyan, Qatar	565.16	2	-1 Valparaiso, Chile	327.31
3	+1 Doha, Qatar	529.85	3	-1 Lyon, France	326.76
4	-2 Dubai, United Arab Emirates	479.90	4	- Beijing, China	296.51
5	+7 Rio de Janeiro, Brazil	388.61	5	- New York, United States	294.05
6	+7 São Paulo, Brazil	361.86	6	- Bucharest, Romania	282.74
7	- Riyadh, Saudi Arabia	324.27	7	+2 Shanghai, China	280.04
8	-2 Sofia, Bulgaria	318.19	8	-1 Zürich, Switzerland	277.84
9	-4 Copenhagen, Denmark	293.48	9	+3 Geneva, Switzerland	275.93
10	+4 Skopje, North Macedonia	283.78	10	+1 New Taipei, Taiwan	270.95
11	-1 Kuwait City, Kuwait	279.90	11	-1 Bangkok, Thailand	270.26
12	+5 Jeddah, Saudi Arabia	265.18	12	-4 Ar-Rayyan, Qatar	269.02

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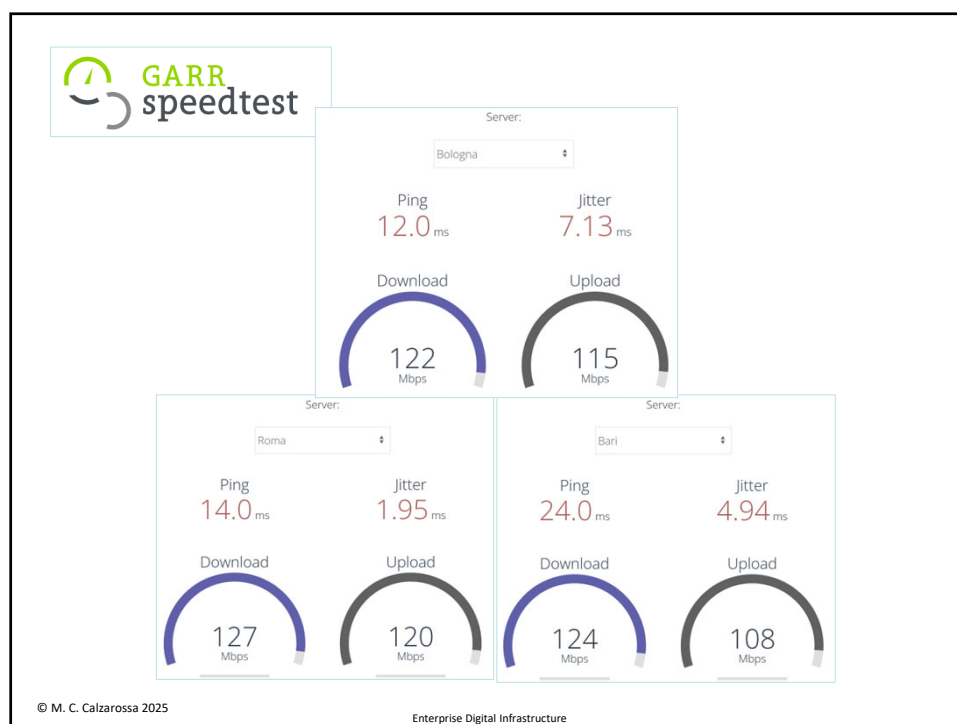
GARR network



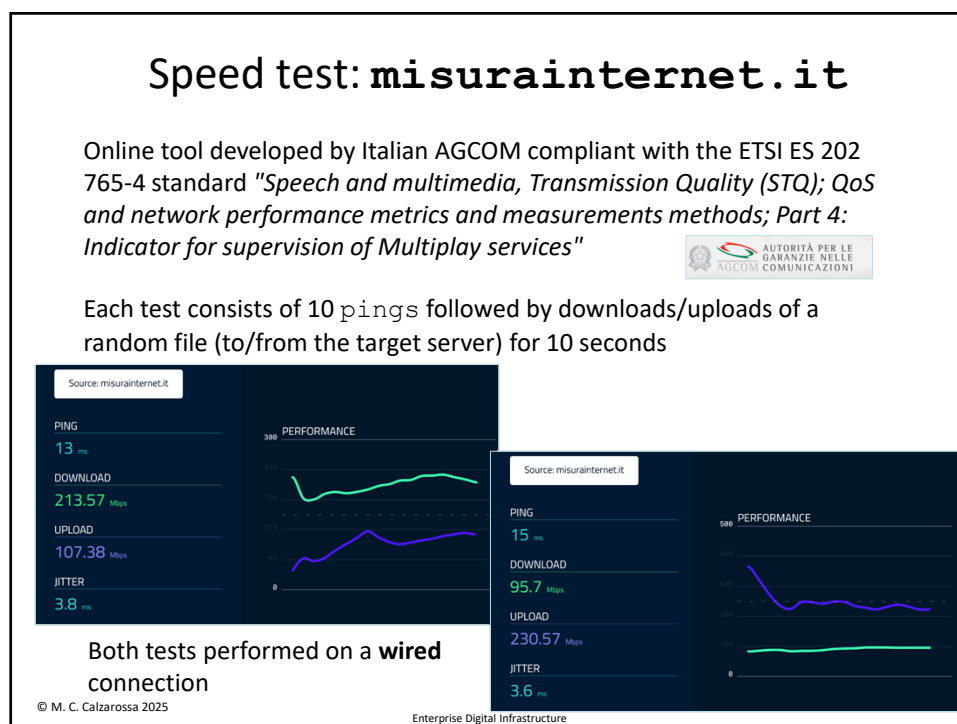
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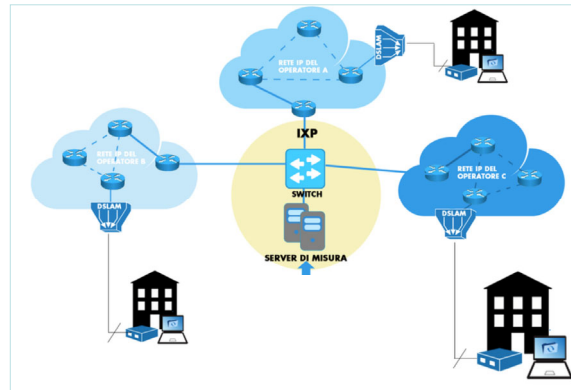


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Active monitoring: NeMeSys

Network Measurement System (NeMeSys) is an open-source software developed by AGCOM to obtain *certified* measurements of the quality of the Internet access offered by ISPs (fixed broadband)

The target is a server located at the *closest* Neutral Access Point



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Source: MisuraInternet

Recap

Active monitoring is a technique used to collect measurements on a *real* technological infrastructure under a *controlled artificial* load/traffic

Load/traffic generators

ping: the vantage point generates ICMP packets towards a target and measures the reachability of the target and the *instantaneous* RTT; the network is seen as a *black-box*

traceroute: the vantage point generates packets towards a target to discover network topology and measures the *instantaneous* RTTs between vantage point and target and between vantage point and intermediate routers

Speed test tools generate traffic from a vantage point towards a target and measure the *maximum* achievable bandwidth

Measurements can vary significantly and are heavily affected by the usage of a shared communication infrastructure that might involve multiple ISPs

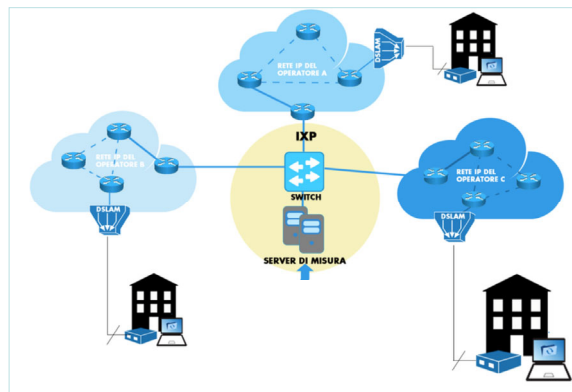
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NeMeSys Architecture

NeMeSys is a benchmark aimed at assessing the performance of Internet connections



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NeMeSys

The load generated from a user device (i.e., vantage point) consists of 10 pings, 20 *uploads* and 20 *downloads* of a test file of variable size

The file size is adjusted to make each test last a minimum amount of time (e.g., 15 minutes)

The measurements collected refer to *speed* (i.e., download and upload bandwidth) and *latency* (i.e., RTT)

Entire testing activities could last up to three days (each test has to be repeated multiple times)

In case measured performance falls below the *contractual performance*, customers can submit *claims* to their providers

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Provider broadband offers



AGCOM publishes online the values of the contractual download bandwidths providers have to guarantee to their customers

TIM	TIM SUPER 10 GIGA (FTTH 10/2 Giga) (Sottoscritto dal 31/07/2023)	Business	FTTH			10 Gbps	500 Mbps	4,5 Gbps	600 Mbps
TIM	TIM WIFI CASA con opzione 10 Giga (FTTH 10/2 Giga)	Consumer	FTTH			10 Gbps	200 Mbps	4 Gbps	300 Mbps

nominal download bandwidth

minimum guaranteed download bandwidth

max download bandwidth

"normal" available download bandwidth

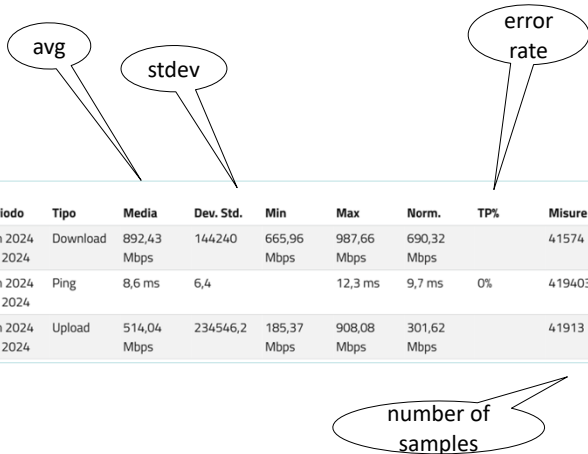
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Provider measurements – 1st semester 2024

[Provider measurements](#) collected every six months from different regional vantage points using NeMeSys (results published online by ACGOM)



Operatore	Profilo	Velocità (down/up)	Territorio	Periodo	Tipo	Media	Dev. Std.	Min	Max	Norm.	TP%	Misure
Fastweb	INTERNET	1 Gbps 300 Mbps	Nazionale	Gen 2024 Giu 2024	Download	892,43 Mbps	144240	665,96 Mbps	987,66 Mbps	690,32 Mbps		41574
Fastweb	INTERNET	1 Gbps 300 Mbps	Nazionale	Gen 2024 Giu 2024	Ping	8,6 ms	6,4		12,3 ms	9,7 ms	0%	419403
Fastweb	INTERNET	1 Gbps 300 Mbps	Nazionale	Gen 2024 Giu 2024	Upload	514,04 Mbps	234546,2	185,37 Mbps	908,08 Mbps	301,62 Mbps		41913

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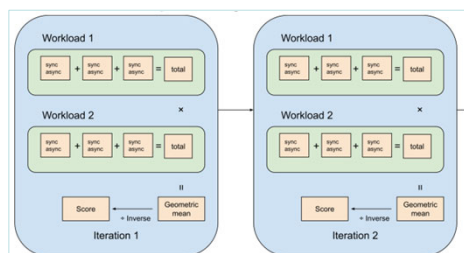
Speedometer 3.0

[Speedometer](#) is a browser benchmark that measures the *responsiveness* of Web applications

It is an open source project developed under the governance of leading companies, i.e., Apple, Google, Microsoft, Mozilla, to promote browser interoperability; it targets web developers

It is based on demo web workloads that realistically simulate user actions, such as download of a complex web page, usage of a To-Do app, reading a news website

The performance score is based on the runtime taken by the user actions of each workload (arithmetic mean of the inverse geometric mean)

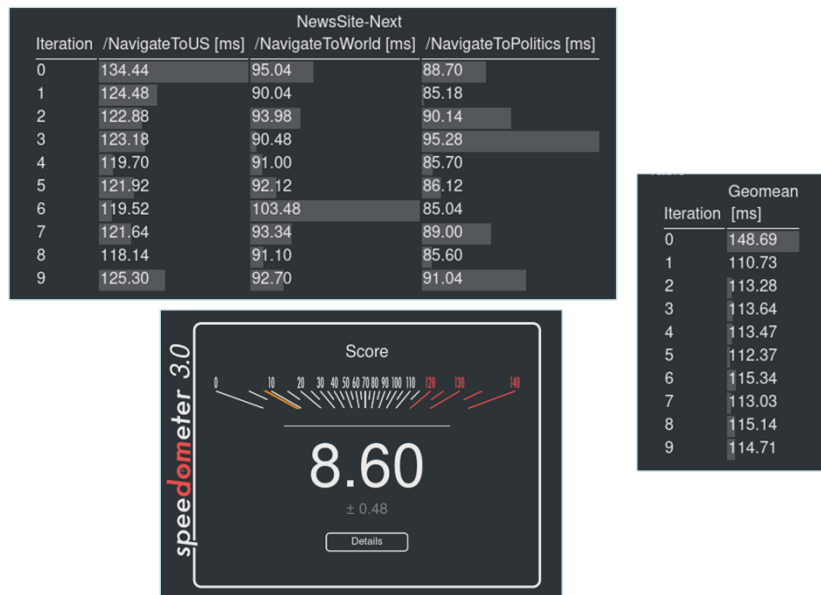


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Results



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Benchmark frameworks

Non-profit organizations promote standard benchmarks to be used as a reference point for comparing hardware and software solutions of different providers/vendors

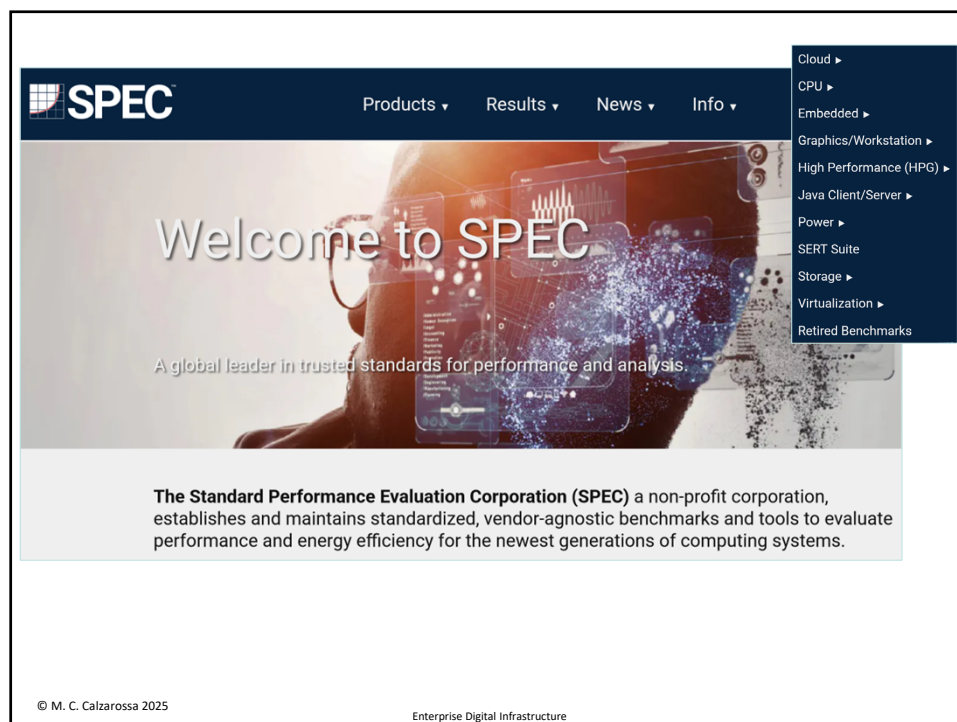
Two main objectives:

1. Creating “good” benchmarks, i.e., benchmarks representative of different application domains
2. Creating a process for reviewing and monitoring the benchmarks, i.e., strict rules associated with the runs of these benchmarks to ensure the *reproducibility* and *repeatability* of the performance results

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SPECworkstation 4.0

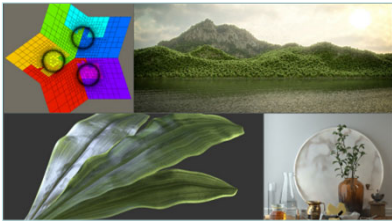
Benchmark released in December 2024 to measure *workstation performance* and ensure professionals make informed decisions on hardware their investments

Free for non commercial usage

Benchmark includes 20+ workloads focused on seven industry verticals (e.g., AI & machine learning, life science, product design) and four hardware subsystems (e.g., CPU, graphics)

Performance is measured in terms of composite scores with respect to a reference system

Composite scores are obtained as a weighted geometric mean of the ratios of individual tests



Component	Specification
CPU	i7-12700 (8C/16T+4E)
Graphics	WX3200 4GB
RAM	2x16GB DDR5 4400
Disk	512GB NVMe M.2 SSD
OS	Windows 11 Professional

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Some results

System Configuration	
Manufacturer	LENOVO
Model	Lenovo ThinkStation PX
CPU	2x INTEL(R) XEON(R) PLATINUM 8592+
Memory	256.00 GB @ 4800 MHz
GPU	NVIDIA RTX 6000 Ada Generation
Display	S24C650 23.4" (1920x1080)
Storage	Drive 1: SAMSUNG MZVL2512HDJD-00BLL 476.94 GB - SCSI Drive 2: KXG8AZNV2T04 LA KIOXIA 1907.73 GB - SCSI Drive 3: KXG8AZNV2T04 LA KIOXIA 1907.73 GB - SCSI Drive 4: KXG8AZNV2T04 LA KIOXIA 1907.73 GB - SCSI Drive 5: KXG8AZNV2T04 LA KIOXIA 1907.73 GB - SCSI
OS	Microsoft Windows 11 Pro for Workstations (22H2)

AI & Machine Learning				
Workload	Reference Result	Measured Result	Unit	SPEC Ratio
Data Science				
Pandas	131.99	150.33	sec	0.88
Scikit-learn	449.17	57.14	sec	7.86
XGBoost	91.50	70.17	sec	1.30

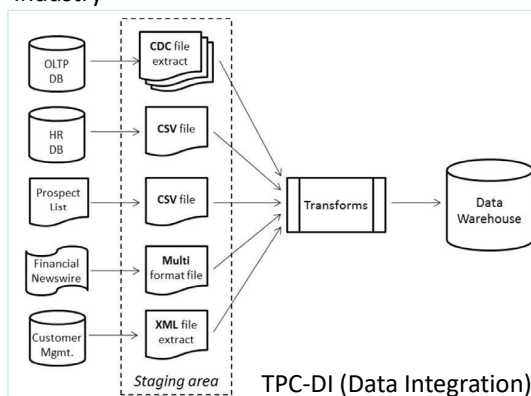
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Transaction Processing Performance Council
focusing on data-centric benchmark standards
and disseminating objective, verifiable data to
industry



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Benchmarks/Results
Transaction Processing
Decision Support
Virtualization
Big Data
Internet of Things
Artificial Intelligence

TPC provides benchmark
specifications and some tool
kits

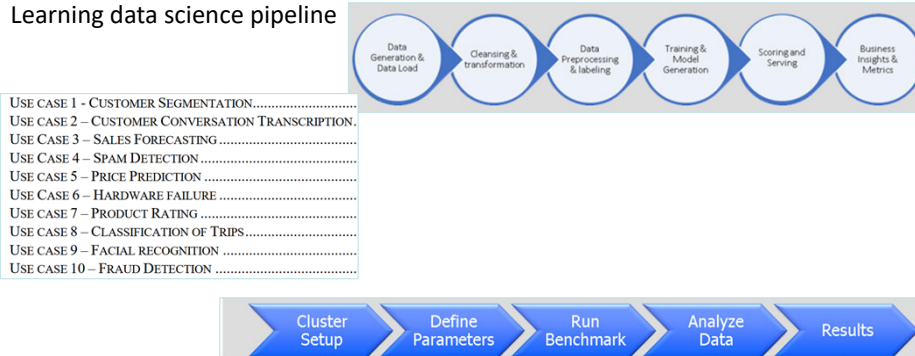
49

TPCx-AI

TPCx-AI is a standard benchmark that measures the performance of a machine learning or data science platform

It emulates the behavior of representative industry AI solutions relevant in current production data centers and cloud environments

A Use Case defines a single problem solved by the Deep Learning and Machine Learning data science pipeline



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Passive monitoring

Passive monitoring techniques and tools aim at measuring the performance and assessing the efficiency of services/infrastructure *while they are operating*

Passive monitoring is based on the *direct observation* of the behavior/operations of a *real* infrastructure under its *actual* workload/traffic

Observability refers to understanding the behavior and the performance of a system through observation

Many diverse techniques/tools for passive monitoring

The choice mainly depends on the objectives of the monitoring project, but also on the target and on the usage of the measurements (i.e., online, offline)

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Recap

Active monitoring includes a large variety of techniques and tools characterized by different levels of complexity

The artificial load or traffic being generated is the *benchmark* used for performance assessment and comparison

Active techniques often used in combination with passive techniques

Passive techniques monitor systems and services while they are operating through direct observation of their behaviors

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Logging facilities

Logging facilities collect measurements of all activities and events (including malfunctioning) generated by applications and services

These facilities are typically integrated into operating systems and applications/services

Compliance with GDPR has to be ensured

Measurements are organized into records stored into log or trace files (e.g., access, audit) according to different formats

Logs are analyzed *offline*

Possible usages: analytics, accounting, auditing, marketing, requested by law

Every activity performed over the Internet is tracked and corresponding data is stored somewhere (even in multiple places)

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Logging facilities: Web logs

Apache access log (stored in a standard format)

```
35.207.5.150 - - [13/Mar/2025:10:50:57 +0100] "GET /ads.txt HTTP/1.1" 404 403 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko; compatible; BW/1.3; rb.gy/qyzae5) Chrome/124.0.0.0 Safari/537.36"
```

```
51.8.102.100 - - [13/Mar/2025:05:06:27 +0100] "GET /publications/PDF/future-internet_preprint.pdf HTTP/1.1" 200 881628 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/131.0.0.0 Safari/537.36; compatible; OAI-SearchBot/1.0; +https://openai.com/searchbot"
```

Apache error log

```
[Thu Mar 06 15:57:13.477269 2025] [autoindex:error] [pid 1991:tid 134574925416128] (13)Permission denied: [client 20.171.207.203:36202] AH01275: Can't open directory for index: /var/www/html/PEG/publications/
[Sat Mar 08 13:28:09.129484 2025] [core:error] [pid 1991:tid 134575080605376] [client 8.219.120.85:36286] AH00244: invalid URI path (/cgi-bin/.%2e/.%2e/.%2e/.%2e/.%2e/.%2e/.%2e/.%2e/.%2e/.%2e/.%2e/bin/sh)
[Sat Mar 08 13:28:15.456026 2025] [core:error] [pid 1991:tid 134574914930368] [client 8.219.120.85:46354] AH00244: invalid URI path (/cgi-bin/%32%65%32%65/%32%65%32%65/%32%65%32%65/%32%65%32%65/%32%65%32%65/%32%65%32%65/%32%65%32%65/bin/sh)
```

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Linux authorization log

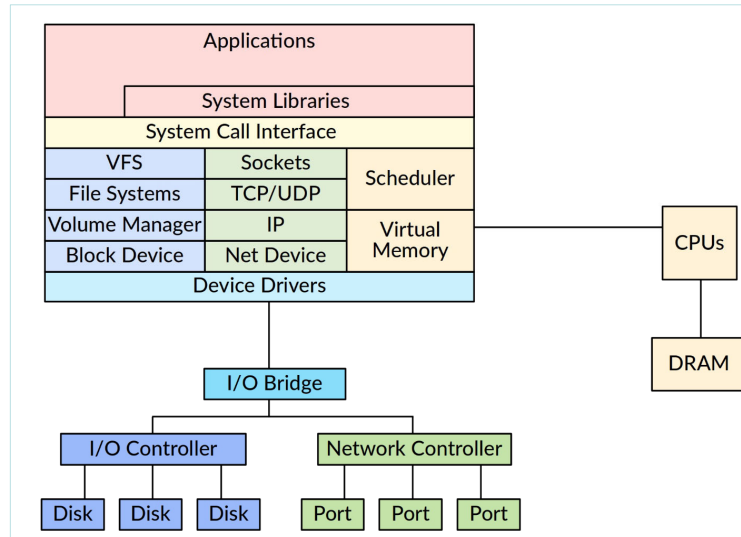
```
2025-03-09T00:02:09.734063+01:00 piccione sshd[1460383]: Invalid user admin from 92.255.85.37 port 34444
2025-03-09T00:02:09.784819+01:00 piccione sshd[1460383]: pam_unix(sshd:auth): check pass; user unknown
2025-03-09T00:02:09.784961+01:00 piccione sshd[1460383]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=92.255.85.37
2025-03-09T00:02:11.534808+01:00 piccione sshd[1460363]: Failed password for root from 218.92.0.114 port 27374 ssh2
2025-03-09T00:02:11.840895+01:00 piccione sshd[1460383]: Failed password for invalid user admin from 92.255.85.37 port 34444 ssh2
2025-03-09T00:02:12.408845+01:00 piccione sshd[1460383]: Connection reset by invalid user admin 92.255.85.37 port 34444 [preauth]
2025-03-09T00:02:12.665692+01:00 piccione sshd[1460446]: Invalid user admin from 92.255.85.37 port 49856
2025-03-09T00:02:12.715154+01:00 piccione sshd[1460446]: pam_unix(sshd:auth): check pass; user unknown
2025-03-09T00:02:12.715310+01:00 piccione sshd[1460446]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=92.255.85.37
2025-03-09T00:02:14.153800+01:00 piccione sshd[1460363]: Failed password for root from 218.92.0.114 port 27374 ssh2
```

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How to measure system components

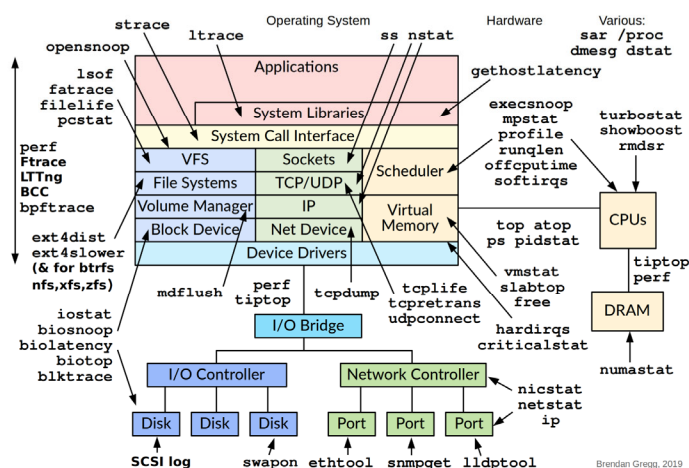


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Linux Observability Tools



Brendam Gregg "Systems Performance: Enterprise and the Cloud, 2nd Edition" (2020)

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Performance monitors

Performance monitors provide measurements about the status and activities of technological infrastructures and services

They are based on the *performance counters*, i.e., data items associated with performance objects, made available by Performance Monitoring Units or by operating systems

They monitor/count *hardware events* and *software events* (e.g., processor clock cycles, cache references, retired instructions, branch misses, page faults, context switches)

Two approaches are applied to collect samples:

- *Periodic event polling* at a specified time interval
- *Continuous event monitoring* such that upon *occurrence* of an event overflows of performance counters trigger interrupts whose handlers enable collecting samples

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Performance monitors (cont'd)

The *Performance Counters for Linux* is a kernel-based subsystem that provides a framework for collecting and analyzing performance data [*perf* command]

The *Microsoft Windows Performance Counters* provide a high-level abstraction layer used for collecting various kinds of system data [*Performance Monitor* command]

The data varies depending on the performance monitoring hardware and the software configuration of the system

System administrators use performance counters to monitor systems and identify performance issues

Software developers use performance counters to examine the behavior and resources used by their applications

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Examples of Linux counters

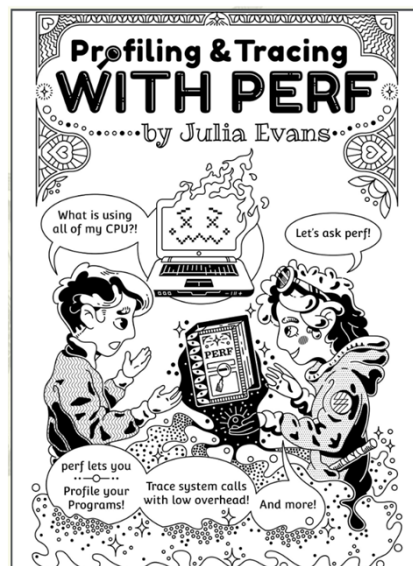
branch-instructions OR branches	[Hardware event]	
branch-misses	[Hardware event]	
bus-cycles	[Hardware event]	
cache-misses	[Hardware cache event]	
cache-references	[Hardware cache event]	
cpu-cycles OR cycles	[Hardware cache event]	
instructions	[Hardware cache event]	
ref-cycles	[Hardware cache event]	
stalled-cycles-backend OR idle-cycles-backend	[Hardware cache event]	
stalled-cycles-frontend OR idle-cycles-frontend	[Hardware cache event]	
alignment-faults	[Software event]	
bpf-output	[Software event]	
cgroup-switches	[Software event]	
context-switches OR cs	[Software event]	
cpu-clock	[Software event]	
cpu-migrations OR migrations	[Software event]	
dummy	[Software event]	
emulation-faults	[Software event]	
major-faults	[Software event]	
minor-faults	[Software event]	
page-faults OR faults	[Software event]	
task-clock	[Software event]	
duration_time	[Tool event]	
user_time	[Tool event]	
system_time	[Tool event]	

cache:	
l1d.replacement	[Counts the number of cache lines replaced in L1 data cache]
l1d_pend_miss.fb_full	[Number of cycles a demand request has waited due to L1D Fill Buffer (FB) unavailability]
l1d_pend_miss.fb_full_periods	[Number of phases a demand request has waited due to L1D Fill Buffer (FB) unavailability]
l1d_pend_miss.l2_stall	[Number of cycles a demand request has waited due to L1D due to lack of L2 resources]
l1d_pend_miss.pending	[Number of L1D misses that are outstanding]
l1d_pend_miss.pending_cycles	[Cycles with L1D load Misses outstanding]

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Linux perf command

Performance counter stats for 'ls -ltrR /':

```

21,117.14 msec task-clock           # 0.486 CPUs utilized
    1,943   context-switches        # 92.011 /sec
      99    cpu-migrations           # 4.688 /sec
     895    page-faults              # 42.383 /sec
29,302,277,401 cycles                # 1.388 GHz
58,162,522,569 instructions          # 1.98 insn per cycle
11,008,601,387 branches              # 521.311 M/sec
  55,845,392 branch-misses           # 0.51% of all branches
TopdownL1                          # 10.5 % tma_backend_bound
                                          # 50.5 % tma_bad_speculation
                                          # 15.1 % tma_frontend_bound
                                          # 15.4 % tma_retiring

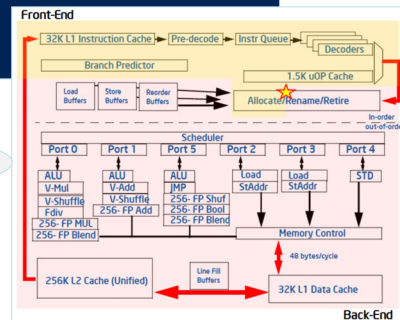
```

43.459838842 seconds time elapsed

5.720055000 seconds user

15.406108000 seconds sys

Top-down Microarchitecture
Analysis - Level 1 [Intel Ice Lake]

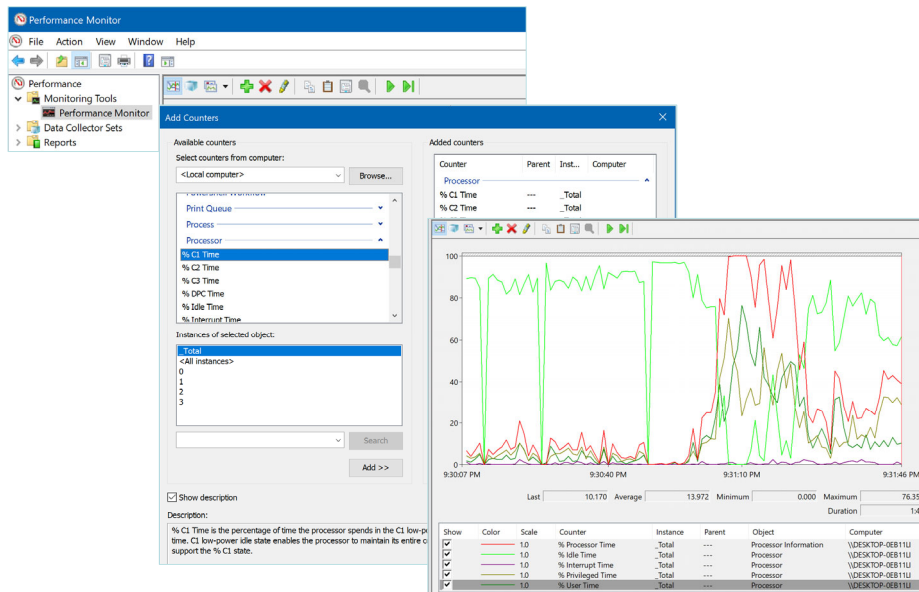


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Microsoft Windows Performance Monitor



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Linux server status: *sysstat* utilities

The *sysstat* utilities are performance monitoring tools for Linux systems

They monitor system performance and usage

These tools are very useful for checking the status of a systems and quickly identifying performance problems (i.e., “crisis” tools)

They collect information about the overall system and of the individual components (e.g., CPU, memory, I/O devices)

“60 seconds tools”: **uptime**, **vmstat**, **mpstat**, **pidstat**, **iostat**, **top**, ...

```
top - 12:44:44 up 7 days, 23:57, 2 users, load average: 0.33, 0.29, 0.32
Tasks: 1049 total, 1 running, 1048 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.2 us, 0.1 sy, 0.0 ni, 99.6 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 257362.9 total, 215202.6 free, 31257.0 used, 14117.2 buff/cache
MiB Swap: 8192.0 total, 8192.0 free, 0.0 used. 226105.9 avail Mem
```

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Performance profilers

Performance profilers are part of the software development ecosystem

They measure the activities and performance of applications/services for identifying (and improving) poorly performing sections of code (*bottlenecks*)

They collect measurements at run time, such as how long a module/function takes to execute, how often it is called, where it is called from, ...

Profilers are often based on code *instrumentation*, i.e., monitoring hooks or probes inserted into the source or binary code of the target application/service

They are characterized by two conflicting requirements: *accuracy* vs *intrusiveness*

Specific profilers for the different programming languages (e.g., gprof, Jprof, JVM monitor) or services (e.g., Mozilla Firefox)

Hybrid solutions exploiting probes and performance counters are often applied

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Python Profilers

Python is equipped with performance profilers that provide useful information to understand where the code spends its time

cProfile is a *deterministic* profiler characterized by a “reasonable” overhead that makes it suitable for profiling long-running programs

Deterministic profiling reflects the fact that all events associated with *function calls*, *function returns*, and *exceptions* are monitored and precise timings are made for the intervals between these events

It is not necessary to manually instrument the code since a hook is automatically associated with each event

The profiler provides a detailed breakdown of the number of calls and the total time spent in each function (with a *millisecond resolution*)

Textual and graphical outputs

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cProfile output

```
30544211 function calls (30200666 primitive calls) in 18.558 seconds

Ordered by: cumulative time
List reduced from 1109 to 23 due to restriction <'features'>

ncalls  tottime  percall  cumtime  percall filename:lineno(function)
      1      0.000      0.000      18.392      18.392 features.py:364(compute_features)
      1      0.002      0.002      18.388      18.388 features.py:369(<listcomp>)
    2304      0.007      0.000      18.385      0.008 features.py:380(_kernel)
    2304      0.016      0.000      18.369      0.008 features.py:382(<listcomp>)
    2304      0.022      0.000      7.432      0.003 features.py:145(compute)
    2304      0.070      0.000      5.800      0.003 features.py:322(compute)
    2304      0.019      0.000      5.384      0.002 features.py:120(compute)
    2304      0.132      0.000      4.798      0.002 features.py:32(hl_envelopes_idx)
    2304      0.032      0.000      4.238      0.002 features.py:162(compute)
    2304      0.542      0.000      2.295      0.001 features.py:56(<listcomp>)
    2304      0.538      0.000      2.263      0.001 features.py:58(<listcomp>)
    2304      0.094      0.000      0.723      0.000 features.py:296(compute)
    2304      0.004      0.000      0.160      0.000 features.py:107(compute)
    2304      0.038      0.000      0.088      0.000 features.py:20(min_max_normalize)
    2304      0.004      0.000      0.004      0.000 features.py:150(<listcomp>)
    2305      0.001      0.000      0.001      0.000 features.py:366(<genexpr>)
      1      0.000      0.000      0.000      0.000 features.py:355(get_feature_names)
      1      0.000      0.000      0.000      0.000 features.py:357(<listcomp>)
      1      0.000      0.000      0.000      0.000 features.py:293(feature_names)
      1      0.000      0.000      0.000      0.000 features.py:142(feature_names)
      1      0.000      0.000      0.000      0.000 features.py:159(feature_names)
      1      0.000      0.000      0.000      0.000 features.py:104(feature_names)
      1      0.000      0.000      0.000      0.000 features.py:319(feature_names)
```

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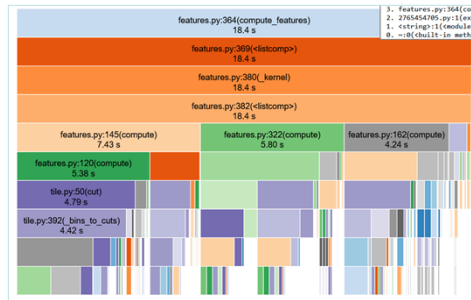
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SnakeViz

SnakeViz is a viewer for Python profiling data that runs as a web application

Two types of diagrams used to represent hierarchical data: *icicle* and *sunburst*

The fraction of time spent in a function is represented by the extent of a visualization element, i.e., width of a rectangle or angular extent of an arc



Icicle plot



Sunburst plot

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Google Chrome profiler

Summary	Bottom-up	Call tree
Range: 0 ms – 1 min		
System	398 ms	
Scripting	284 ms	
Rendering	280 ms	
Painting	66 ms	
Loading	49 ms	
Total	60,705 ms	

Summary	Bottom-up	Call tree	Event log
Aa (*)	ab	Filter	No grouping
Self time	Total time	Activity	
135.3 ms	18.7 %	261.5 ms	36.1 %
105.8 ms	14.6 %	105.8 ms	14.6 %
4.7 ms	0.6 %	71.9 ms	9.9 %
58.0 ms	8.0 %	58.0 ms	8.0 %
42.0 ms	5.8 %	47.7 ms	6.6 %
36.5 ms	5.0 %	45.8 ms	6.3 %
32.0 ms	4.4 %	32.0 ms	4.4 %
8.4 ms	1.2 %	24.8 ms	3.4 %
19.8 ms	2.7 %	19.8 ms	2.7 %
14.9 ms	2.1 %	14.9 ms	2.1 %
14.6 ms	2.0 %	14.6 ms	2.0 %
11.1 ms	1.5 %	11.1 ms	1.5 %

Summary	Bottom-up	Call tree	Event log
Aa (*)	ab	Filter	≥ 1 ms
Start time	Self time	Total time	Activity
0.8 ms	2.4 ms	38.5 ms	Evaluate script
2.9 ms	36.1 ms	36.1 ms	Profiling overhead
13,562.7 ms	6.5 ms	6.5 ms	Parse HTML
13,693.5 ms	1.4 ms	1.4 ms	Parse stylesheet
13,698.7 ms	3.5 ms	3.5 ms	Parse stylesheet
13,709.1 ms	1.4 ms	1.4 ms	Parse stylesheet
13,719.8 ms	0.5 ms	5.9 ms	Evaluate script
13,720.0 ms	1.3 ms	1.3 ms	Compile script

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Service/application monitoring

Performance debugging/diagnosis refer to the identifications of inefficient components of the applications (i.e., *bottlenecks*) and the development of appropriate optimization strategies

Two possible approaches to monitor the performance of an application:

- *Performance monitoring*: measurements being collected are associated with *performance counters* and provide a *low-level* view of the behavior and performance characteristics of the application
- *Performance profiling*: measurements being collected are associated with the *individual sections* of the code and provide a detailed *high-level* view of the behavior and performance characteristics of the application

Measurements collected at run time are analyzed offline

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Recap

Passive monitoring techniques vary depending on the objectives and on the target

Logging facilities collect information about activities/events associated with systems, services and applications

Performance monitors focus on the behavior and efficiency of hardware and software components (mostly used by system administrators)

Profilers help software developers to investigate the behavior of their applications and debug their performance

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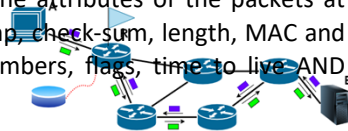
Sniffers

Sniffers are monitoring tools for *network diagnostic* that collect measurements at a *microscopic* level by *eavesdropping* the traffic flowing on wired or wireless networks

They run on any device whose Network Interface Card is set to promiscuous mode (i.e., the NIC captures and stores all traffic flowing on the network)

The vantage point determines the portion of network being monitored

Measurements collected by sniffers refer to the attributes of the packets at each level of the protocol stack, e.g., timestamp, check-sum, length, MAC and IP addresses, source and destination port numbers, flags, time to live AND payloads ... **compliance with GDPR**



Depending on the objective of the monitoring project, measurements can be used either online or offline (i.e., live capture and offline analysis)

Serious issues concerning security and privacy (e.g., open WiFi)

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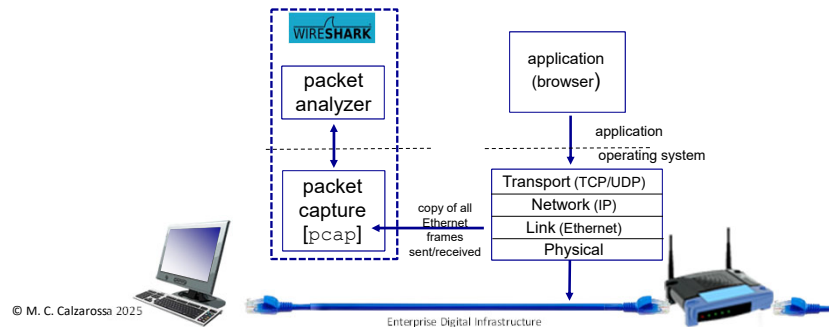
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Sniffer implementation

Wireshark is the de facto standard used for network sniffing (and analysis) within enterprises and institutions

TShark is a terminal oriented version of Wireshark designed for capturing and displaying packets when an interactive user interface is not necessary/available

The **libpcap** is a portable C/C++ library developed within the **tcpdump** project and used to capture network traffic directly from NICs



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Wireshark capture

The screenshot shows the Wireshark interface with a list of captured packets. The 'Packet List' pane shows several packets, with packet 4014 selected. The 'Packet Details' pane shows the structure of packet 4014, which is a TCP segment. The 'Packet Bytes' pane shows the raw data of the selected packet.

No.	Time	Source	Destination	Protocol	Length	Info
4010	9.449810	10.87.152.123	193.204.40.97	TCP	54	60875 → 3128 [ACK] Seq=589 Ack=2947 Win=131328 Len=0
4011	9.450003	10.87.152.123	193.204.40.97	TCP	54	60875 → 3128 [FIN, ACK] Seq=589 Ack=2947 Win=131328 Len=0
4012	9.450132	142.251.209.8	10.87.152.123	TCP	66	443 → 60879 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
4013	9.450214	10.87.152.123	142.251.209.8	TCP	54	60879 → 443 [ACK] Seq=1 Ack=1 Win=131072 Len=0
4014	9.451704	10.87.152.123	142.251.209.8	TCP	1466	60879 → 443 [ACK] Seq=1 Ack=1 Win=131072 Len=1412 [TCP PDU reassembled in 4015]
4015	9.451704	10.87.152.123	142.251.209.8	TLSv1.3	551	Client Hello (SNI=www.googletagmanager.com)
4016	9.452585	10.87.152.123	193.204.40.230	HTTP	515	GET /wp-includes/js/jquery/jquery-migrate.min.js?ver=1.4.1 HTTP/1.1
4017	9.453101	10.87.152.123	193.204.35.27	DNS	84	Standard query 0x2756 A www.googletagmanager.com
4018	9.454510	193.204.40.230	10.87.152.123	HTTP	238	HTTP/1.1 200 OK (text/css)

Frame 4014: 1466 bytes on wire (11728 bits), 1466 bytes captured (11728 bits) on interface \Device\NPF_{6D0E9EDB-F001-4A51-ABE5-10C...}

Ethernet II, Src: Microsoft_cb:7e:ae (bc:83:85:cb:7e:ae), Dst: Fortinet_09:00:1b (00:09:0f:09:00:1b)

Internet Protocol Version 4, Src: 10.87.152.123, Dst: 142.251.209.8

Transmission Control Protocol, Src Port: 60879, Dst Port: 443, Seq: 1, Ack: 1, Len: 1412

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Transmission Control Protocol, Src Port: 60879, Dst Port: 443, Seq: 1, Ack: 1, Len: 1412

- Source Port: 60879
- Destination Port: 443
- [Stream index: 30]
- [Stream Packet Number: 4]
- > [Conversation completeness: Incomplete, DATA (15)]
- [TCP Segment Len: 1412]
- Sequence Number: 1 (relative sequence number)
- Sequence Number (raw): 521152425
- [Next Sequence Number: 1413 (relative sequence number)]
- Acknowledgment Number: 1 (relative ack number)
- Acknowledgment number (raw): 3726978039
- 0101 = Header Length: 20 bytes (5)
- ▼ Flags: 0x010 (ACK)
 - 000. = Reserved: Not set
 - ...0 = Accurate ECN: Not set
 - 0... = Congestion Window Reduced: Not set
 -0.. = ECN-Echo: Not set
 -0. = Urgent: Not set
 -1 = Acknowledgment: Set
 -0 = Push: Not set
 -0 = Reset: Not set
 -0. = Syn: Not set
 -0 = Fin: Not set

Internet Protocol Version 4, Src: 10.87.152.123, Dst: 142.251.209.8

- 0100 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 1452
- Identification: 0x8ae8 (35560)
- > 010. = Flags: 0x2, Don't fragment
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 128
- Protocol: TCP (6)
- Header Checksum: 0x678d [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 10.87.152.123
- Destination Address: 142.251.209.8
- [Stream index: 45]

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Wireshark capture: content

Frame 4767: 784 bytes on wire (6272 bits), 784 bytes captured (6272 bits) on interface \Device\NPF_{608E9EDB-F001-4A51-ABE5-10C908F3BF40}, id 0

- > Ethernet II, Src: TplinkTechno_95:d0:68 (98:48:27:95:d0:68), Dst: Microsoft_cb:7e:ae (bc:83:85:cb:7e:ae)
- > Internet Protocol Version 4, Src: 193.204.40.230, Dst: 192.168.1.103
- > Transmission Control Protocol, Src Port: 80, Dst Port: 50793, Seq: 201548, Ack: 6230, Len: 730
- ▼ Hypertext Transfer Protocol
 - > HTTP/1.1 200 OK\r\n
 - Date: Tue, 18 Mar 2025 19:55:43 GMT\r\n
 - Server: Apache/2.2.16 (Debian)\r\n
 - Last-Modified: Thu, 09 Feb 2017 08:27:40 GMT\r\n
 - Accept-Ranges: bytes\r\n
 - > Content-Length: 446\r\n
 - Keep-Alive: timeout=15, max=86\r\n
 - Connection: Keep-Alive\r\n
 - Content-Type: image/png\r\n
 - X-Pad: avoid browser bug\r\n
 - \r\n
 - [Request in frame: 4763]
 - [Time since request: 0.078100000 seconds]
 - [Request URI: /wp-content/themes/DiplunipIng/images/pagination.png]
 - [Full request URI: <http://webing.unipv.eu/wp-content/themes/DiplunipIng/images/pagination.png>]
 - File Data: 446 bytes
- ▼ Portable Network Graphics
 - PNG Signature: 89504e470d0a1a0a
 - > Image Header (IHDR)
 - > Textual data (tEXt)
 - > Image data chunk (IDAT)
 - > Image Trailer (IEND)

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Traffic analysis: expert information

Wireshark · Expert Information · WiFi			
Severity	Summary	Group	Protocol
Error	Malformed Packet (Exception occurred)	Malformed	IEEE 802.11
Warning	ACKed segment that wasn't captured (common at capture start)	Sequence	TCP
Warning	Connection reset (RST)	Sequence	TCP
Warning	D-SACK Sequence	Sequence	TCP
Note	ACK to a TCP keep-alive segment	Sequence	TCP
Note	TCP keep-alive segment	Sequence	TCP
Note	Partial Acknowledgement of a segment	Sequence	TCP
Note	Time To Live	Sequence	IPv4
Note	This frame is a (suspected) spurious retransmission	Sequence	TCP
Note	This frame undergoes the connection closing	Sequence	TCP
Note	This frame initiates the connection closing	Sequence	TCP
Note	The acknowledgment number field is nonzero while the ACK flag is not set	Protocol	TCP
Note	Duplicate ACK	Sequence	TCP
Note	This frame is a (suspected) retransmission	Sequence	TCP
Chat	Connection finish (FIN)	Sequence	TCP
Chat	This legacy_version field MUST be ignored. The supported_versions exte...	Deprecated	TLS
Chat	Connection establish acknowledge (SYN+ACK)	Sequence	TCP
Chat	Connection establish request (SYN)	Sequence	TCP

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Traffic analysis: endpoints

Wireshark · Endpoints · WiFi

Endpoint Settings

☐ Name resolution

☐ Limit to display filter

Copy

Map

Protocol

☒ Ethernet

☐ FC

☐ FDDI

☐ IEEE 802.11

☐ IEEE 802.15.4

☒ IPv4

☒ IPv6

Filter list for specific type

Ethernet · 343

IPv4 · 83

IPv6 · 8

TCP · 61

UDP · 113

Address	Port	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
4.207.247.137	443	8	883 bytes	3	524 bytes	5	359 bytes
10.87.148.122	7680	5	330 bytes	0	0 bytes	5	330 bytes
10.87.152.123	49426	2	121 bytes	1	55 bytes	1	66 bytes
10.87.152.123	56593	3	381 bytes	2	152 bytes	1	229 bytes
10.87.152.123	56594	3	381 bytes	2	152 bytes	1	229 bytes
10.87.152.123	56648	2	121 bytes	1	55 bytes	1	66 bytes
10.87.152.123	60606	33	3 kB	13	1 kB	20	2 kB
10.87.152.123	60610	12	2 kB	5	825 bytes	7	706 bytes
10.87.152.123	60644	4	282 bytes	2	108 bytes	2	174 bytes
10.87.152.123	60684	14	1 kB	6	634 bytes	8	693 bytes
10.87.152.123	60727	14	1 kB	6	630 bytes	8	689 bytes
10.87.152.123	60756	19	5 kB	8	4 kB	11	2 kB
10.87.152.123	60758	11	6 kB	4	3 kB	7	4 kB
10.87.152.123	60822	2,319	2 MB	465	188 kB	1,854	2 MB
10.87.152.123	60854	5	330 bytes	5	330 bytes	0	0 bytes
10.87.152.123	60855	5	330 bytes	5	330 bytes	0	0 bytes

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Traffic analysis: Packet lengths

Wireshark · Packet Lengths · WiFi								
Topic / Item	Count	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Start
▼ Packet Lengths	7348	549.57	42	1514	0.3040	100%	5.0500	6.181
0-19	0	-	-	-	0.0000	0.00%	-	-
20-39	0	-	-	-	0.0000	0.00%	-	-
40-79	4142	57.02	42	79	0.1714	56.37%	1.0200	11.365
80-159	327	98.84	80	157	0.0135	4.45%	0.2900	5.305
160-319	154	250.68	161	317	0.0064	2.10%	0.1300	7.346
320-639	205	474.06	324	638	0.0085	2.79%	0.2300	9.434
640-1279	162	940.54	642	1265	0.0067	2.20%	0.1600	6.219
1280-2559	2358	1476.50	1285	1514	0.0976	32.09%	4.3300	6.181
2560-5119	0	-	-	-	0.0000	0.00%	-	-
5120 and greater	0	-	-	-	0.0000	0.00%	-	-

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Microsoft Packet Monitor

Packet Monitor (**pktmon**) is a built-in sniffer for Microsoft Windows environments (e.g., MS Azure) aimed at *network diagnostics*

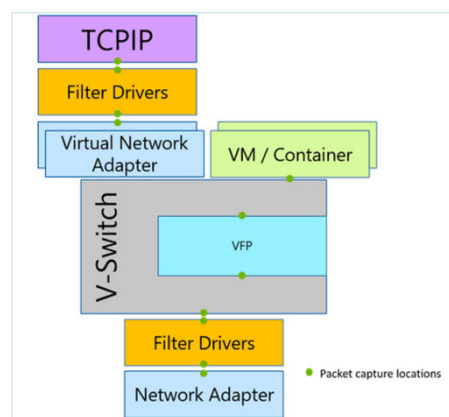
It is very useful in virtualized scenarios (e.g., Software Defined Networks) because it provides visibility within the entire network stack

It can be used for packet capture, drop detection, filtering and counting

Data is saved into Event Trace Log (ETL) files (that can be converted into **pcap** format and analyzed using Wireshark)

No GUI

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pktmon

```
PS C:\WINDOWS\system32> pktmon
pktmon <command> [OPTIONS | help]
Advanced packet capture and event collection.
```

```
Commands
  filter      Manage packet filters.
  list        List packet processing components.

  start       Start packet capture and event collection.
  stop        Stop data collection.
  status       Query current status.
  unload       Unload PktMon driver.
```

```
counters      Display current packet counters.
reset          Reset packet counters to zero.
```

```
etl2txt        Convert log file to text format.
etl2pcap       Convert log file to pcap format.
hex2pkt        Decode packet in hex format.
```

```
help           Show help text for pktmon s
Example: pktmon s
```

Marvell AVASTAR Wireless-AC Network Controller									
Id Name		Counter	Direction	Packets	Bytes	Direction	Packets	Bytes	

12 Marvell AVASTAR Wireles...		Drops	Rx	1	86	Tx	29	1,742	
		Upper	Rx	47,037	43,431,141	Tx	35,486	6,016,562	

40 WFP Native Filter		Lower	Rx	47,037	43,431,141	Tx	35,486	6,016,562	
		Upper	Rx	47,037	43,431,141	Tx	35,486	6,016,562	
39 Npcap Packet Driver (NP...		Lower	Rx	47,037	43,431,141	Tx	35,486	6,016,562	
		Upper	Rx	47,037	43,431,141	Tx	35,486	6,016,562	
38 Virtual WiFi Filter Driver		Lower	Rx	47,037	43,431,141	Tx	35,486	6,016,562	
		Upper	Rx	47,037	43,431,141	Tx	35,486	6,016,562	
37 Native WiFi Filter Driver		Lower	Rx	47,037	43,431,141	Tx	35,486	6,016,562	
		Upper	Rx	47,031	42,491,191	Tx	35,480	5,377,070	
36 Npcap Packet Driver (NP...		Lower	Rx	47,031	42,491,191	Tx	35,480	5,377,070	
		Upper	Rx	47,031	42,491,191	Tx	35,480	5,377,070	
35 QoS Packet Scheduler		Lower	Rx	0	0	Tx	35,480	5,377,070	
		Upper	Rx	0	0	Tx	35,480	5,377,070	
34 WFP 802.3 Filter		Lower	Rx	47,031	42,491,191	Tx	35,480	5,377,070	
		Upper	Rx	47,031	42,491,191	Tx	35,480	5,377,070	

73 TCP/IP		Lower	Rx	47,030	42,491,105	Tx	35,509	5,378,812	
71 LLTDIO		Lower	Rx	47,031	42,491,191	Tx	0	0	
69 NDISUIO		Lower	Rx	47,031	42,491,191	Tx	0	0	

86 HTTP (192.168.1.104)		Lower	Rx	2	959	Tx	1	2,351	

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Measurement platforms

Measurement platforms refer to the technological infrastructures used to collect performance measurements

Probes, i.e., measurement points, available at strategic locations within access and backbone networks and behind residential gateways

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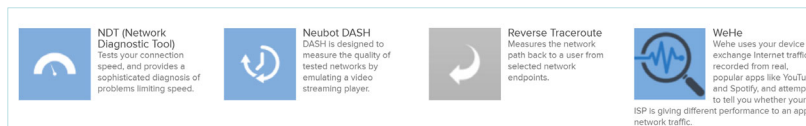
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Measurement Lab

[Measurement Lab](#) (M-Lab) is a non-profit large project supported by organizations, educational institutions and private companies aimed at advancing Internet research by *empowering consumers with useful information about their Internet performance*

The platform provides measurement tools and widely distributed servers to support *active* measurements



All tools hosted by M-lab *open source*

Data collected by M-Lab global measurement platform are made *openly available* (including the IP addresses that users use to connect to the test)

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M-Lab infrastructure map



Each physical M-Lab “site” consists of 3-4 servers and one switch, connected directly to an upstream provider

Virtual sites include 1 or more virtual servers proportionate to demand

Multiple sites are located in large metro area to obtain diversity in routes

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RIPE Atlas

[RIPE Atlas](#) is a measurement platform deployed by the RIPE Network Coordination Centre for collecting Internet measurements

It employs a global network of probes (i.e., small, USB-powered hardware devices connected to an Ethernet port on a router) that measure Internet connectivity and reachability



Probes perform active measurements, including ping, traceroute, DNS, SSL/TLS and NTP

The collected data is aggregated and made publicly available

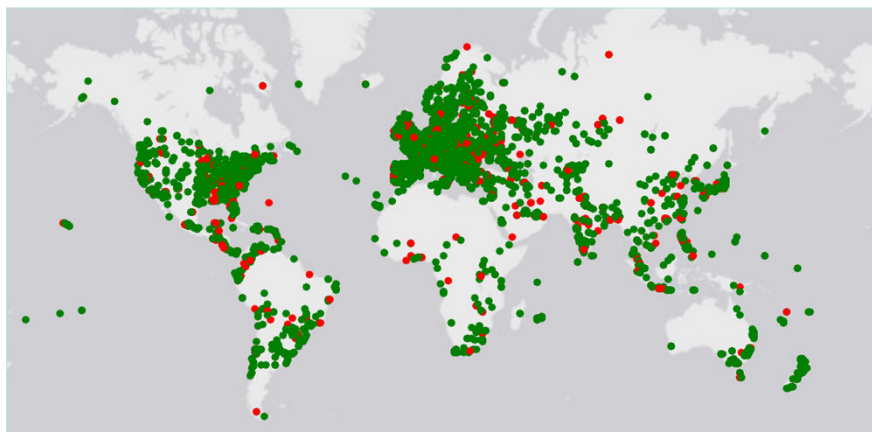
Users who host a probe can also use the entire RIPE Atlas network to conduct customised measurements

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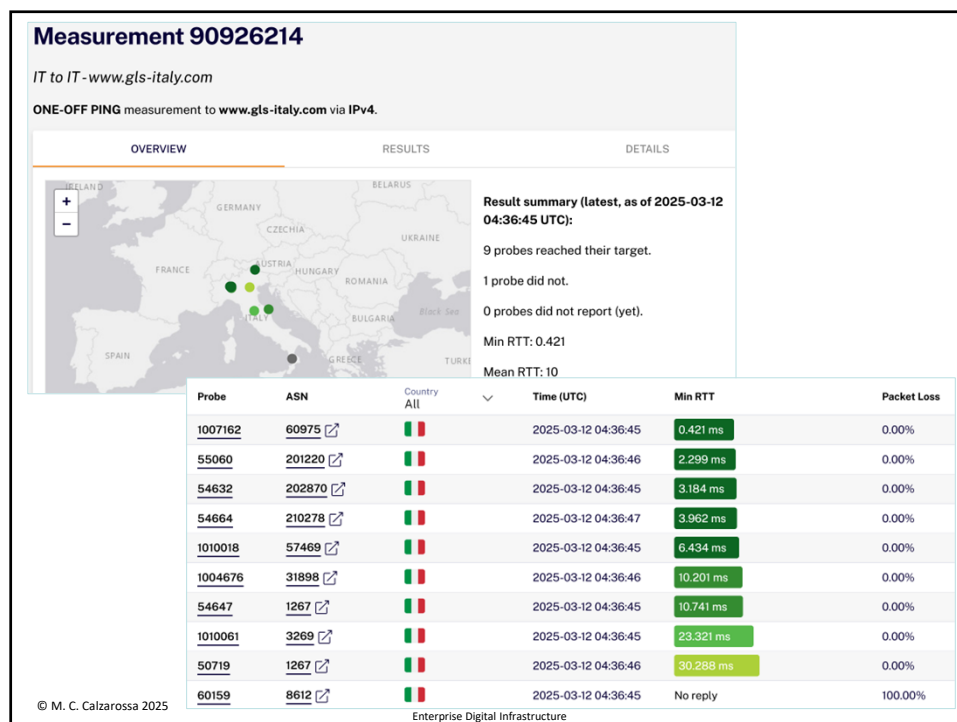
RIPE Coverage



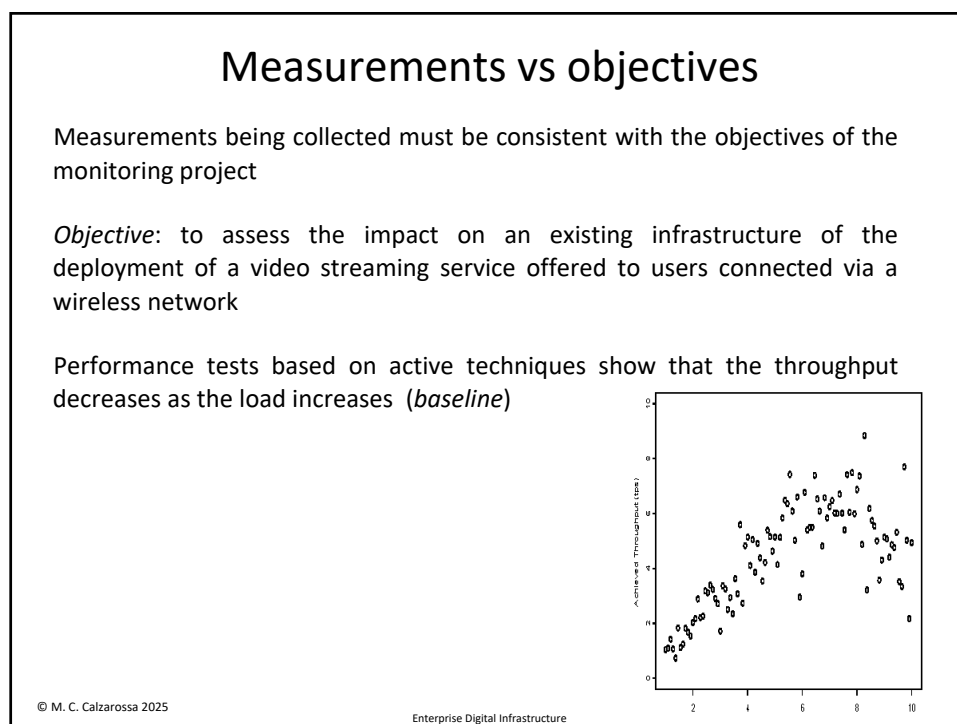
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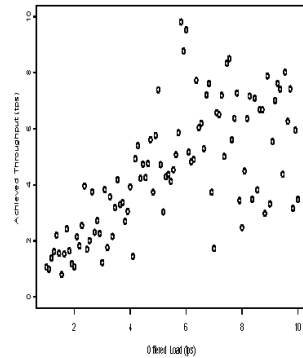
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Measurements vs objectives (cont'd)

Conjecture: the wireless network is the bottleneck

The provider upgrades the infrastructure by adding two Access Points

Performance tests show that the throughput improves but not as much as it should



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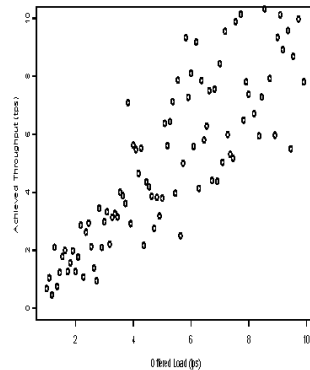
90

Measurements vs objectives

More detailed investigations (i.e., passive techniques combined with active techniques) that consider the entire infrastructure show that:

1. Packet loss rate is very small
2. RAM of the server is overutilized (i.e., bottleneck)

After the RAM upgrade, the throughput increases with the load as expected



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Monitoring challenges/recommendations

Monitoring overhead and intrusiveness might affect the operating conditions and performance of technological infrastructures and services

Both aggressive and conservative monitoring activities should be avoided

An appropriate tradeoff between the amount of measurements to be collected and the “cost” associated with the measurement process has to be achieved

Unexpected effects might appear when monitoring activities are not performed in dedicated environments (e.g., speed tests, virtualized environments)

It is recommended to repeat the experiments multiple times to ensure the quality of the measurements

Legal, privacy and ethical issues (e.g., GDPR, “acceptable” usage policy) should also be considered very seriously

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Measurement quality

Measurement quality is of paramount importance since measurements are at the basis of many decision processes

It is necessary to assess the accuracy and precision of the selected monitoring tools (e.g., slow packet sniffing)

Changing insignificant and seemingly irrelevant aspects of the infrastructure used in the monitoring experiments can introduce a *bias* in the measurements (e.g., geographic coverage)

Representativeness and validity of the measurements should be ensured to avoid the risk of ignoring significant although *rare* phenomena

Sampling techniques should be applied with particular care (e.g., mice and elephant phenomenon)

Domain knowledge is a fundamental prerequisite for any monitoring activity

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